

Teaching Students to Use AI as a Learning Partner

Materials, process, and what we learned from classroom evidence

Brainstorm

Build

Revise

Core idea

AI helps students think, plan, check, and revise. Students stay responsible for choices, voice, and final work.

Module Context

Three-session sequence

1. **Lesson 1: Brainstorm and Plan** – choose a topic, check the CS connection, generate a high-level guide.
2. **Lesson 2: Build Step by Step** – use the guide to build project sections with AI support.
3. **Lesson 3: Revise and Validate** – finish, compare to the plan, revise, and explain AI use.

Product menu

Slideshow, poster/one-pager, booklet, or simple site.

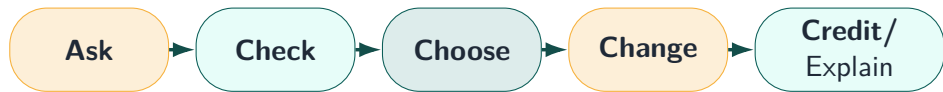
Materials Used

Material	What it does	Link
Main lesson packet	Defines the full 3-session AI partner project, goals, norms, product menu, and assessment frame.	packet PDF
Lesson 1 folder	Holds setup, student directions, prompt cards, rubric, exit ticket, and slides for brainstorming.	lesson1 folder
Lesson 2 folder	Holds setup, student directions, rubric, exit ticket, and slides for building from the guide.	lesson2 folder
Transcript anchor	Shows how Lesson 1 went in class and what needed adjustment for Lesson 2.	Apr. 10 transcript
Official references	Ground safety, AI literacy, CS standards, and inclusion choices.	bibliography

Official Reference Anchors

- ▶ **MagicStudent**: teacher-managed student AI access.
- ▶ **MagicSchool privacy and security**: safe, monitored classroom use.
- ▶ **ISTE AI in Education**: AI literacy and responsible classroom integration.
- ▶ **UNESCO generative AI guidance**: human agency, age-appropriate use, and checking AI output.
- ▶ **CSTA K-12 Standards**: CS-aligned problem solving, tools, and responsible computing.

AI Partner Mini-Framework



Why it matters

The routine makes AI use observable: students ask for help, check the response, choose what fits, change it, and explain their own decision.

Lesson 1: Brainstorm and Plan

Goal

Students leave with a topic, CS connection, product choice, and high-level guide.

Tool flow

Idea Generator first: generate CS-connected topic ideas.

AI Assistant second: check, narrow, and organize one topic.

Student process

1. Pick an interest.
2. Generate possible CS topics.
3. Choose one manageable topic.
4. Check CS fit.
5. Generate a guide with 3–5 parts.

Lesson 1 Materials

Material	Purpose	Link
AI setup	Teacher-facing configuration and tool roles for Idea Generator and AI Assistant.	setup
AI directions	Student-facing workflow, approved prompts, and copying/checking rule.	directions
Prompt cards	Copy-friendly prompt page for student access.	prompt cards
Student slides	Classroom scaffolding for mission, rules, tool separation, workflow, and prompts.	slides PDF
Rubric + exit ticket	Captures topic ownership, CS connection, planning with AI, safety, and readiness.	rubric / exit ticket

What the Transcript Showed for Lesson 1

Classroom observations from April 10, 2026

- ▶ Students needed explicit support for Google Classroom, duplicate tabs, MagicSchool login, and tool navigation.
- ▶ Prompt copying had to be modeled slowly: select, copy, paste, replace the blank, then submit.
- ▶ Student interests like space, soccer, animals, games, and cats were useful entry points.
- ▶ The CS connection had to stay visible so topics did not become general-interest projects.
- ▶ Many students reached a topic or project idea, but not all completed the full guide.

Transcript-Based Adjustment

What worked

- ▶ Clear tool separation helped.
- ▶ Copyable prompts reduced typing load.
- ▶ Student choice increased engagement.
- ▶ Teacher modeling made AI use safer.

What needed improvement

- ▶ Login and tab routines took real time.
- ▶ Some students needed help choosing from AI output.
- ▶ Lesson 2 needed a recovery checkpoint before building.
- ▶ Success criteria needed to remain visible during work time.

Lesson 2: Build Step by Step

Goal

Students use their topic and guide to build project sections, not restart brainstorming.

Tool flow

AI Assistant: recover/clarify/expand one guide part.

Project tool: build the actual slideshow, poster, booklet, or site.

Student process

1. Recover topic and guide.
2. Build Part 1.
3. Build Part 2.
4. Build Part 3.
5. Check against guide and CS goal.

Lesson 2 Materials

Material	Purpose	Link
AI setup	Teacher-facing setup for a single bounded AI tool and project-building workflow.	setup
AI directions	Student-facing directions for using AI as a build partner.	directions
Student slides	Recovery checkpoint, tool split, workflow, prompts, worked example, and midpoint check.	slides PDF
Rubric	Checkpoint rubric for using the plan, step-by-step execution, CS accuracy, safe AI use, and Lesson 3 readiness.	rubric
Exit ticket	Captures what students built, how AI helped, what they changed, and next steps.	exit ticket

Why Lesson 2 Starts With Recovery

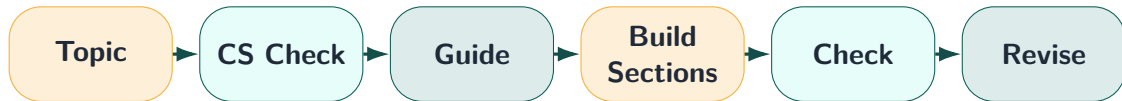
Instructional reason

The transcript showed that Lesson 1 produced strong engagement and many project ideas, but the full guide was uneven across students.

Lesson 2 response

- ▶ Confirm or rebuild topic, product type, CS connection, and guide.
- ▶ Move quickly from recovery into building.
- ▶ Use AI Assistant only for one section at a time.
- ▶ Keep the project tool as the place where the real work is built.

Student Workflow Across the Module



Teacher move

Keep students moving from AI conversation into visible student work.

CS Standards and Responsible AI

CS learning

- ▶ Use digital tools for a defined purpose.
- ▶ Break a project into smaller parts.
- ▶ Evaluate output from a computing system.
- ▶ Organize information into a plan.
- ▶ Communicate a digital product to an audience.

Responsible AI learning

- ▶ No private information in prompts.
- ▶ AI output can be wrong.
- ▶ Students choose what to keep or change.
- ▶ Final work must show student voice.
- ▶ Teacher monitors classroom AI use.

Inclusive Strategies

- ▶ Visual prompt frames and projected workflows reduce working-memory load.
- ▶ Copyable prompts reduce typing and spelling barriers.
- ▶ Students can talk, draw, point, or rehearse before typing.
- ▶ Partner roles support reading directions and checking ideas.
- ▶ Chunked checkpoints make progress visible.
- ▶ Exit tickets capture learning even when the final product is not finished.

What We Learned

Instructional takeaways

- ▶ AI lessons need as much tool-routine scaffolding as content scaffolding.
- ▶ Prompt access matters: students need fast ways to copy, paste, and replace blanks.
- ▶ Student choice works best when bounded by a clear CS connection.
- ▶ Lesson 2 should not assume Lesson 1 ended perfectly; recovery is part of good design.
- ▶ Assessment should capture student decisions, not just AI output.

Closing Frame

**AI is the partner.
Students are the thinkers.**

What to look for in the materials

The strongest evidence of learning is when students can explain what they asked, what they checked, what they kept, what they changed, and why their project still meets the goal.